



Draft genome sequences of 25 *Salmonella enterica* serovar Agona strains isolated from poultry and associated food products harbouring multiple antibiotic resistance genes

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ARTICLE INFO

Article history:

Received 24 October 2021

Revised 7 February 2022

Accepted 20 February 2022

Available online 25 February 2022

Editor: Prof Marco R Oggioni

Keywords:

Whole genome sequencing

Salmonella enterica

Agona

ST13

Non-typhoidal *Salmonella*

NTS

ABSTRACT

Objectives: Antimicrobial-resistant livestock-associated *Salmonella enterica* serovar Agona infection poses a significant public health threat worldwide. The present study aimed to identify antibiotic resistance genes in livestock-associated *S. Agona* strains isolated from chickens and associated food products (meat and eggs) in Pakistan via whole-genome sequencing.

Methods: The genomic DNAs of *S. Agona* strains (n=25) were sequenced using an Illumina MiSeq platform. The generated reads were trimmed and de novo assembled using CLC Genomics Workbench v.7. The draft genomes were annotated using the National Centre for Biotechnology Information (NCBI) Prokaryotic Genome Annotation Pipeline and were characterised by multilocus sequence typing (MLST). The antimicrobial-resistance genes (acquired and chromosomal mutations), extrachromosomal plasmids and *Salmonella* pathogenicity islands were predicted using ResFinder and CARD, PlasmidFinder and SPIFinder, respectively.

Results: The genome size of *S. Agona* ranges from 4.9 to 5.1 Mb with 52.1% GC contents. The strains belong to ST13 and harbour several antibiotic-resistance genes, including *aac* (6')-Iaa, *aadA1*, *aadA2*, *bla* OXA-10, *qnrS1*, *cmlA*, *floR*, *tet(A)*, *dfrA12* and point mutations in *gyrB*, *gyrA*, *ParC* conferring antibiotic resistance to fluoroquinolones. The strains also contain several plasmids and *Salmonella* pathogenicity islands.

Conclusion: This study reports draft genomes of multidrug-resistant *S. Agona* from Pakistan isolated from chickens and associated food products. The data may help with understanding the antimicrobial resistance mechanisms and transmission dynamics of this serovar in poultry and associated food products and their possible transmission to humans.

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1. Introduction

Salmonella enterica is one of the most prevalent bacteria that cause foodborne illnesses around the world. Non-typhoidal

Salmonella (NTS) is a major public health concern worldwide due to its disease-causing potential in different hosts, including livestock and humans. Approximately 153 million cases of gastroenteritis and 57 000 deaths are attributed to NTS infections globally each year [1]. In recent years, *Salmonella* Agona has been ranked among top 10 frequently isolated NTS serovars in China and among top 20 frequently isolated serovars in the United States [2].

Salmonella Agona has been associated with several foodborne outbreaks in the last few decades globally. A wide range of foods and animal products, including infant milk formula, meat, chicken and eggs, has been linked to these outbreaks [3]. In recent years, multidrug resistant *S. enterica* serovar Agona has been posing a

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Table 1
Genomic characterization of 25 *Salmonella enterica* serovar Agona isolates from poultry and associated food products

Sequence ID	Laboratory ID	Source	Accession no.	MLST	Resistance genes	Point mutation related to fluoroquinolone resistance*	Detected plasmid Plasmid	Genome			
								Size (kb)	Length bp	Contig L50	GC contents(%)
FMBL1	FML1	Fecal	SAMN18914338	13	<i>FosA7, golS, aadA1, aph(3')-IIa, ARR-2,blaOXA10,cmlA1,dfrA14-floR-,QnrS1,tet(A),</i>	<i>acrB: p.L40P, gyrB: p.T717N, gyrA: p.D87Y, parC: p. T57S</i>	<i>IncX1</i>	41.2	4,993,443	6	52.5
FMBL2	FML2	Fecal	SAMN18923267	13	<i>arr-2, aadA1, aph(3')-IIa-,ARR-2,blaOXA-10, AAC(6')-ly, cmlA1,dfrA14,floR,QnrS1,tet(A)</i>	<i>parC: p.T57S, gyrA: p. D87Y</i>	<i>IncX1</i>	41.2	4,926,698	9	52.1
FMBL3	FML3	Fecal	SAMN19012489	13	<i>aph(3')-IIa, floR-,QnrS1,tet(A), FosA7</i>	<i>parC: p.S469A, gyrB: p.T717N</i>	<i>IncX1</i>	41.2	4,955,047	6	52.2
FMBL4	OML1	Liver	SAMN19012512	13	<i>aph(3')-IIa, floR,QnrS1,tet(A)</i>	<i>parC: p.T57S</i>	<i>IncX1</i>	41.2	4,974,858	6	52.1
FMBL5	FML4	Fecal	SAMN19012488	13	<i>aph(3')-IIa, floR,QnrS1,tet(A)</i>	<i>gyrA: p. D87Y</i>	<i>IncX1</i>	41.2	4,915,615	11	52.1
FMBL6	FML5	Fecal	SAMN19012496	13	<i>aadA1-laph (3')-IIa, ARR-2, blaOXA-10, ampC1, cmlA1, dfrA14,floR,QnrS1,tet(A)</i>	<i>gyrA: p.D87Y, parC: p. T57S, gyrB: p. T717N</i>	<i>IncX1</i>	41.2	4,909,034	6	52.3
FMBL8	FML6	Fecal	SAMN19012490	13	<i>aadA2, dfrA12, floR, QnrS1, sul1, tet(A)</i>	<i>gyrA: p.D87Y, parC: p. S469A</i>	<i>IncFIB(K)</i>	105.1	4,976,042	7	52.3
FMBL9	FML6	Fecal	SAMN19012491	13	<i>aadA2, dfrA12, floR, QnrS1, sul2, tet(A)</i>	<i>gyrA: p.D87Y, parC: p. S469A</i>	<i>IncFIB(K)</i>	105.1	5,002,609	6	52.3
FMBL11	FML7	Fecal	SAMN19012531	13	<i>aadA1, aph(3')-IIa, ARR-2,blaOXA10,cmlA1,dfrA14, floR,QnrS1,tet(A),</i>	<i>parC: p.T57S, gyrB: p. T717N</i>	<i>IncX1</i>	41.2	4,986,701	6	52.1
FMBL12	FML8	Meconium	SAMN19094650	13	<i>aph(3')-IIa, floR,QnrS1,tet(A)</i>	<i>gyrA: p.D87Y, parC: p.S469A</i>	<i>IncX1</i>	41.2	4,999,937	8	52.1
FMBL14	FML9	Meconium	SAMN19012532	13	<i>aadA1, aph(3')-IIa, ARR-2,blaOXA10,cmlA1,dfrA14,floR, QnrS1,tet(A)</i>	<i>parC: p.T57S, gyrB: p. T717N</i>	<i>IncX1</i>	41.2	4,909,310	6	52.1
FMBL17	FML10	Fecal	SAMN19012533	13	<i>aadA2, dfrA12, floR-l, QnrS1, sul3, tet(A)</i>	<i>gyrA: p.D87Y, parC:p.T57S</i>	<i>IncFIB(K)</i>	105.1	4,961,596	6	52.2
FMBL21	FML11	Fecal	SAMN19012494	13	<i>aadA1, aph(3')-IIa, ARR-2,blaOXA10,cmlA1,dfrA14,floR, QnrS1,tet(A)</i>	<i>acrB: p.L40P, gyrB: p.T717N, gyrA: p.D87Y, parC: p. T57S</i>	<i>IncX1</i>	41.2	4,898,392	7	52.1
FMBL24	FML12	Fecal	SAMN19012492	13	<i>aadA1, aph(3')-IIa, ARR-2,blaOXA-10,cmlA1,dfrA14,floR,QnrS1,tet(A)</i>	<i>acrB: p.L40P, gyrB: p.T717N, gyrA: p.D87Y, parC: p.T57S</i>	<i>IncX1</i>	41.2	4,972,437	9	52.2

(continued on next page)

Table 1 (continued)

Sequence ID	Laboratory ID	Source	Accession no.	MLST	Resistance genes	Point mutation related to fluoroquinolone resistance*	Detected plasmid	Genome			
								Size (kb)	Length bp	Contig L50	GC contents(%)
FMBL25	FML13	Faecal	SAMN19012498	13	<i>aadA2, dfrA12, floR, QnrS1, sul1, tet(A)</i>	<i>parC: p.T57S, gyrB: p.T717N</i>	<i>IncFIB(K)</i>	105.1	4,969,890	6	52.5
FMBL31	OML2	Liver	SAMN19012505	13	<i>aadA2, dfrA12, floR, QnrS1, sul1, tet(A)</i>	<i>parC: p.T57S, gyrB: p. T717N</i>	<i>IncFIB(K)</i>	105.1	5,117,900	7	52.2
FMBL32	PML1	Meat	SAMN19012503	13	<i>aadA2, dfrA12, floR, QnrS1, sul1, tet(A)</i>	<i>parC: p.T57S, gyrB: p. T717N</i>	<i>IncFIB(K)</i>	105.1	5,033,818	6	52.2
FMBL33	PML2	Egg	SAMN19012504	13	<i>aadA1, aph(3')-IIa, ARR-2, blaOXA10, cmlA1, dfrA14, floR, QnrS1, tet(A)</i>	<i>gyrA: p.D87Y, parC: p.S469A</i>	<i>IncX1</i>	41.2	4,949,795	6	52.1
FMBL34	OML3	Liver	SAMN19012501	13	<i>aadA1, aph(3')-IIa, ARR-2, blaOXA10, cmlA1, dfrA14, floR, QnrS1, tet(A)</i>	<i>gyrA: p.D87Y, parC: p. S469A</i>	<i>IncX1</i>	41.2	5,081,832	6	52.1
FMBL35	PML3	Egg	SAMN19012502	13	<i>aadA1, aph(3')-IIa, ARR-2, blaOXA10, cmlA1, dfrA14, floR, QnrS1, tet(A)</i>	<i>parC: p.T57S, gyrB: p. T717N</i>	<i>IncX1</i>	41.2	5,080,457	6	52.1
FMBL36	OML4	Ovaries	SAMN19012510	13	<i>aadA1, aph(3')-IIa, ARR-2, blaOXA10, cmlA1, dfrA14, floR, QnrS1, tet(A)</i>	<i>gyrA: p.D87Y, parC: p. S469A</i>	<i>IncX1</i>	41.2	5,117,406	6	52.2
FMBL37	PML4	Meat	SAMN19012500	13	<i>aadA1, aph(3')-IIa, ARR-2, blaOXA10, cmlA1, dfrA14, floR, QnrS1, tet(A)</i>	<i>acrB: p.L40P, gyrB: p.T717N, gyrA: p.D87Y, parC: p. T57S</i>	<i>IncX1</i>	41.2	4,904,926	8	52.1
FMBL38	PML5	Meat	SAMN19012508	13	<i>aadA2, aph(3')-IIa, ARR-2, blaOXA10, cmlA1, dfrA14, floR, QnrS1, tet(A)</i>	<i>parC: p.T57S, gyrB: p.T717N</i>	<i>IncX1</i>	41.2	5,071,095	6	52.1
FMBL39	PML6	Meat	SAMN19012511	13	<i>aadA1, aph(3')-IIa, ARR-2, blaOXA10, cmlA1, dfrA14, floR, QnrS1, tet(A)</i>	<i>parC: p.T57S, gyrB: p. T717N</i>	<i>IncX1</i>	41.2	5,060,453	6	52.2
FMBL40	OML5	Liver	SAMN19012509	13	<i>aph(3')-IIa, floR, QnrS1, tet(A)</i>	<i>gyrA: p.D87Y, parC: p.S469A</i>	<i>IncX1</i>	41.2	4,934,878	6	52.3

MLST, multilocus sequence typing.

* Sequences compared with GenBank strain ATCC 9150 and deposited in GenBank with accession no. NC006511.

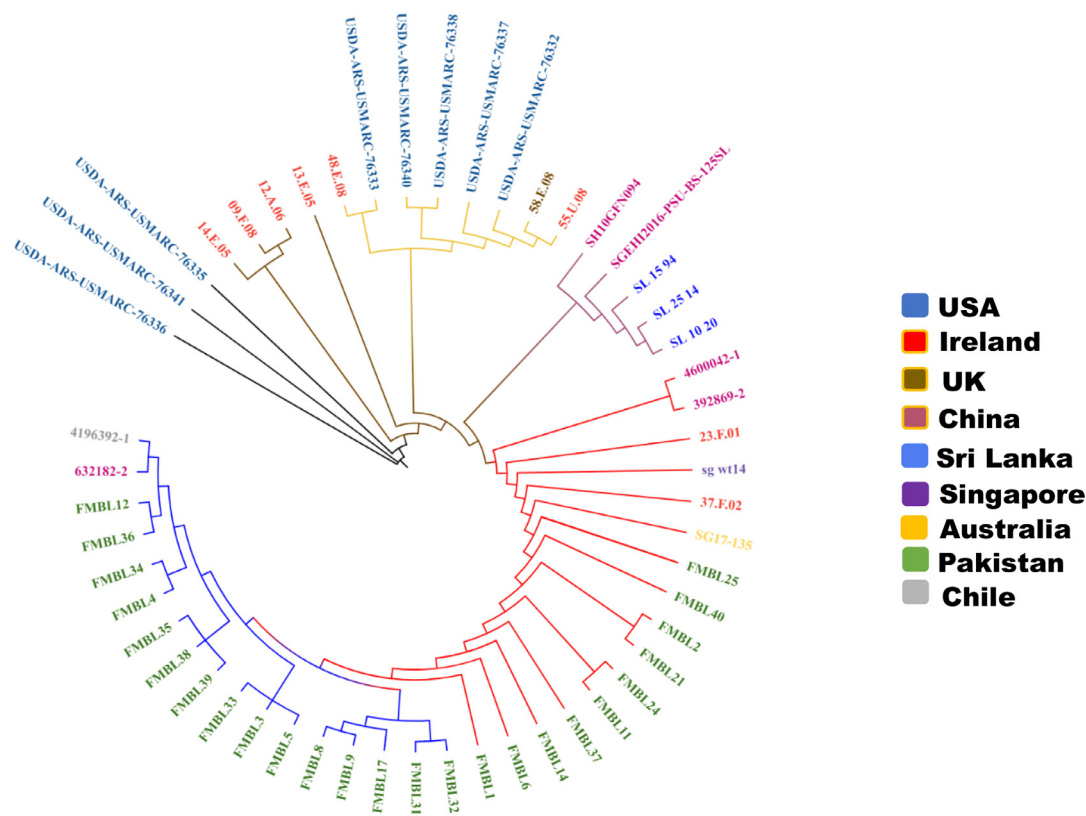


Figure 1. SNP based Phylogenetic tree showing the clonal relationship between 25 *Salmonella enterica* serovar Agona and additional *Salmonella* Agona strains of international origin downloaded from NCBI. A total of 2002253 positions was found in all analyzed genomes. The origin of the strain is represented by the color of the sequence IDs, and the clades are represented by the color of branch length. The colors, red and blue represent the clades of our isolates with other international strains.

major hazard to human and animal health [4]. Whole-genome sequencing of *S. Agona* is essential to understand its pathogenicity and antimicrobial resistance and comparative genomic analysis of this serovar. Here, we report draft genomes of 25 multidrug resistant *S. Agona* isolated from poultry and its associated food products in Pakistan in 2018 [5].

2. Material and methods

The phenotypic characteristics and antibiotic susceptibility of *S. Agona* strains isolated from poultry faeces, organs and associated food products have been described in our previous study [5]. Bacterial stocks were prepared and shipped to the sequencing centre, where DNA extraction was performed using the Roche MagNA Pure 24 platform. Library preparation was performed using Illumina DNA prep (previously known as FLEX), and sequence was run using MiSeq Reagent Kits v3 (600 cycle). Initial QC was performed using QC metrics on BaseSpace, and in-depth QC was performed on BioNumerics. Sequences were analysed and uploaded to NCBI using BioNumerics 7.6 software. The genomes were annotated with the automated NCBI Prokaryotic Genome Annotation Pipeline (PGAP). The strains were characterised by in silico multilocus sequence typing (MLST) using MLST 2.0 (<https://cge.cbs.dtu.dk/services/MLST/>). The antimicrobial resistance genes (acquired and chromosomal mutations) and *Salmonella* pathogenicity islands were identified using ResFinder 3.2 (<https://cge.cbs.dtu.dk/services/ResFinder/>), CARD (<https://card.mcmaster.ca/>), and SPIFinder 1.0 (<https://cge.cbs.dtu.dk/services/SPIFinder/>), respectively. The SNP-based phylogenetic analysis was performed by submitting the genomes to CSI phylogeny 1.4 online server (<https://cge.cbs.dtu.dk/services/CSIPhylogeny/>). The pipeline was run with the default parameters. The raw reads were uploaded, and the analysis was sub-

mitted with the parameters listed below: a minimum depth of 10 reads at SNP positions, a minimum relative depth of 10% at SNP positions, and a minimum distance between SNPs of 10 bp. The assembled genomes were uploaded and aligned with a reference genome (accession no. CVM28336).

3. Results and discussion

The de novo assembly of raw reads generated and average number of 57.4 ± 7.1 contigs. The average size of *S. Agona* draft genome was 4.98 ± 0.06 Mb, which is in agreement with previously published *S. Agona* genome size [6]. The average G + C content is $52.1 \pm 0.12\%$, and multilocus sequence typing (<https://cge.cbs.dtu.dk/services/MLST/>) showed that the isolate belongs to sequence type ST13. The resistome of *S. Agona* isolates consists of genes responsible for resistance to aminoglycosides (*aac* [6']-*laa*, *aadA1* and *aadA2*), β -lactam (*bla* OXA-10), fluoroquinolones (*qnrS1*), phenicols (*cmlA1* and *floR*), sulphonamides (*sul1*, *sul2* and *sul3*), tetracyclines (*tet*[A] and *tet*[M]) and trimethoprim (*dfrA12*). The nonsynonymous point mutations in *gyrA* (p.D87Y), *gyrB* (p.T171N), *acrB*: p.L40P and *ParC* (T57S), which was reported as a genetic marker for this serotype and conferring resistance to nalidixic acid and ciprofloxacin (Table 1). Among 25 *S. Agona* isolates, two plasmid replicons, IncX1 (41.2kb) with 100% similarity with *pESBL3311-IncX1* (accession no. MW390547) and *IncFIB(K)* (105.1kb) with 100% similarity with plasmid *pIncFIBK* (accession no. CP053773), were predicted through Plasmid finder. Several *Salmonella* pathogenicity islands, including SPI-1, SPI-2, SPI-3, SPI-4, SPI-5, SPI-8, SPI-9 and C63PI, were identified in the genomes. According to an SNP-based phylogenetic tree, our strains were closely related to each other, and they form a clade with isolates from China, Chile, Singapore and Australia (Fig. 1).

In conclusion, this study is the first to report the genomic characteristics of multidrug-resistant *S. Agona* isolates from Pakistan. These findings may help researchers better understand the multidrug resistance mechanisms and transmission dynamics of *Salmonella* infection in animals and humans.

Data availability

Data for this whole-genome sequencing project has been deposited in NCBI under the Bio project accession number PRJNA271470. The accession numbers of the draft genome sequence of *S. Agona* are listed in Table 1.

Acknowledgements

The authors would like to thank the Maryland Department of Health Laboratories Administration and FDA for contributing with sequencing of isolates.

Funding

FDA-LFFM 5 U19 FD007084 grant is supported by the FDA to sequence environmental isolates.

Competing interests

None declared.

Ethical approval

Not required.

References

- [1] Gal-Mor O. Persistent infection and long-term carriage of typhoidal and non-typhoidal *Salmonellae*. *Clin Microbiol Rev* 2019;32(1) e00088–18. doi:[10.1128/CMR.00088-18](https://doi.org/10.1128/CMR.00088-18).
- [2] Ferrari RG, Rosario DK, Cunha-Neto A, Mano SB, Figueiredo EE, CA Conte-Junior. Worldwide epidemiology of *Salmonella* serovars in animal-based foods: a meta-analysis. *Appl Environ Microbiol* 2019;85(14) e00591–19. doi:[10.1128/AEM.00591-19](https://doi.org/10.1128/AEM.00591-19).
- [3] Authority EFS, Prevention ECfD, Control. The European Union summary report on trends and sources of zoonoses, zoonotic agents and food-borne outbreaks in 2017. *EFSA J* 2018;16(12):e05500. doi:[10.2903/j.efsa.2018.55004](https://doi.org/10.2903/j.efsa.2018.55004).
- [4] Hu Y, He Y, Wang Y, Fanning S, Cui S, Chen Q, et al. Serovar diversity and antimicrobial resistance of non-typhoidal *Salmonella enterica* recovered from retail chicken carcasses for sale in different regions of China. *Food Control* 2017;81:46–54. doi:[10.1016/j.foodcont.2017.05.0315](https://doi.org/10.1016/j.foodcont.2017.05.0315).
- [5] Siddique A, Azim S, Ali A, Andleeb S, Ahsan A, Imran M, et al. Antimicrobial resistance profiling of biofilm forming non typhoidal *Salmonella enterica* isolates from poultry and its associated food products from Pakistan. *Antibiotics* 2021;10(7):785. doi:[10.3390/antibiotics100707856](https://doi.org/10.3390/antibiotics100707856).
- [6] Zhang S, Yin Y, Jones MB, Zhang Z, Deatherage Kaiser BL, Dinsmore BA, et al. *Salmonella* serotype determination utilizing high-throughput genome sequencing data. *J Clin Microbiol* 2015;53(3):1685–92. doi:[10.1128/JCM.00323-15](https://doi.org/10.1128/JCM.00323-15).